

A Continuous-Dose Extension of the Causal Cluster Variance Estimator: Theory and Proofs

Formal Econometric Derivation

May 25, 2026

Abstract

We extend the fixed-effect Causal Cluster Variance (CCV) estimator proposed by Abadie, Athey, Imbens, and Wooldridge (2023) from a binary treatment framework to continuous-dose treatments. By defining a cluster-level continuous variation parameter based on Frisch-Waugh-Lovell residualized treatment vectors, we derive a continuous-treatment CCV weight that forms a strictly convex combination of the Eicker-Huber-White robust variance and the Liang-Zeger clustered variance estimators. We formally prove that this formula is a strict mathematical generalization of the AAIW binary weight. Furthermore, we define the higher-moment location-scale assumptions required for asymptotic optimality and demonstrate the estimator’s boundary behavior using the Adverse Effect Wage Rate (AEWR) “bite” as a censored continuous application.

1 Setup and The Residual-Maker Projection

Let $g = 1, \dots, G$ index counties or county-like clusters, and let $i = 1, \dots, n_g$ index observations within county g (for example, county-year, worker-county-year, or farm-county-year observations). Let Y_{gi} be the observed outcome, and let D_{gi} be the continuous treatment intensity. An applied example of D_{gi} is the continuous Adverse Effect Wage Rate (AEWR) “bite” intensity:

$$D_{gi} = \max\{0, \log(\text{AEWR}_{gi}) - \log(w_{gi}^0)\}. \quad (1)$$

Let Z_{gi} collect all covariates and fixed effects (e.g., county fixed effects, year fixed effects, control variables) other than D_{gi} . The regression model of interest is:

$$Y_{gi} = \beta D_{gi} + Z'_{gi}\theta + u_{gi}. \quad (2)$$

Stacking the data for the full sample of size $N = \sum_{g=1}^G n_g$, we write the system as $Y = D\beta + Z\theta + u$.

Let $M_Z = I - Z(Z'Z)^{-1}Z'$ be the residual-maker matrix after projecting onto the covariates and fixed effects. By the Frisch-Waugh-Lovell (FWL) theorem, we define the residualized variables as:

$$\tilde{Y} = M_Z Y, \quad \tilde{D} = M_Z D, \quad \tilde{u} = M_Z u. \quad (3)$$

The Ordinary Least Squares (OLS) estimator for the treatment parameter is given by:

$$\hat{\beta} = \frac{\tilde{D}'\tilde{Y}}{\tilde{D}'\tilde{D}}. \quad (4)$$

Let $S = \sum_{g=1}^G \sum_{i=1}^{n_g} \tilde{D}_{gi}^2$. Substituting $\tilde{Y} = \tilde{D}\beta + \tilde{u}$, the estimation error expands to:

$$\hat{\beta} - \beta = S^{-1} \sum_{g=1}^G \sum_{i=1}^{n_g} \tilde{D}_{gi} \tilde{u}_{gi}. \quad (5)$$

Defining the observation-level score as $\psi_{gi} = \tilde{D}_{gi} \tilde{u}_{gi}$ and the county-level score as $\Psi_g = \sum_{i=1}^{n_g} \psi_{gi}$, we rewrite the estimation error as:

$$\hat{\beta} - \beta = S^{-1} \sum_{g=1}^G \Psi_g. \quad (6)$$

The variance of this object is the primary target of our estimation strategy.

2 Robust vs. Clustered Variance Estimators

The difference between standard error regimes is rooted in the treatment of the cross-products within equation (6). The Eicker-Huber-White (robust) variance estimator treats the observation-level scores independently:

$$\hat{V}_{\text{robust}} = S^{-2} \sum_{g=1}^G \sum_{i=1}^{n_g} \hat{\psi}_{gi}^2. \quad (7)$$

Conversely, the standard Liang-Zeger cluster-robust variance estimator aggregates scores at the cluster level before squaring:

$$\hat{V}_{\text{cluster}} = S^{-2} \sum_{g=1}^G \left(\sum_{i=1}^{n_g} \hat{\psi}_{gi} \right)^2. \quad (8)$$

Lemma 1. *The difference between the cluster-robust and heteroskedasticity-robust variance estimators is solely comprised of within-cluster score cross-products.*

Proof. Expanding the squared sum in $\widehat{V}_{\text{cluster}}$ yields:

$$\begin{aligned}\widehat{V}_{\text{cluster}} &= S^{-2} \sum_{g=1}^G \left[\sum_{i=1}^{n_g} \widehat{\psi}_{gi}^2 + \sum_{i \neq j} \widehat{\psi}_{gi} \widehat{\psi}_{gj} \right] \\ &= \widehat{V}_{\text{robust}} + S^{-2} \sum_{g=1}^G \sum_{i \neq j} \widehat{\psi}_{gi} \widehat{\psi}_{gj}.\end{aligned}$$

□

According to the design-based framework established by Abadie, Athey, Imbens, and Wooldridge (2023) [1], the inclusion of the cross-product term $\sum_{i \neq j} \widehat{\psi}_{gi} \widehat{\psi}_{gj}$ should be governed by the fraction of the treatment assignment variation that occurs across clusters versus within clusters.

3 The Continuous-Treatment Causal Cluster Variance Estimator

In the AAIW (2023) framework with a binary treatment $W_{gi} \in \{0, 1\}$ and cluster fixed effects, the treatment variation parameter is based strictly on the within-cluster variance of the binary indicator, $\bar{W}_g(1 - \bar{W}_g)$.

To generalize this logic to a continuous treatment vector D , we define the continuous analog using the FWL-residualized treatment vector. Let the residualized treatment variance within cluster g be:

$$\widehat{\Omega}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} \widetilde{D}_{gi}^2. \quad (9)$$

If panel lengths n_g are highly unequal and the target estimand is unit-weighted rather than equal-county-weighted, we define the sum-of-squares metric:

$$\widehat{S}_g = \sum_{i=1}^{n_g} \widetilde{D}_{gi}^2. \quad (10)$$

Theorem 1 (Continuous CCV Estimator). *Let $q \in [0, 1]$ represent the population sampling fraction. The continuous-dose causal cluster variance weight is defined as:*

$$\widehat{\lambda}_D = 1 - q \frac{\left(G^{-1} \sum_{g=1}^G \widehat{\Omega}_g \right)^2}{G^{-1} \sum_{g=1}^G \widehat{\Omega}_g^2}. \quad (11)$$

The final variance estimator is the resulting convex combination:

$$\widehat{V}_{CCV,D} = \widehat{\lambda}_D \widehat{V}_{cluster} + (1 - \widehat{\lambda}_D) \widehat{V}_{robust}. \quad (12)$$

For unbalanced panel specifications targeting the unit-level Average Treatment Effect, the estimator substitutes $\widehat{S}_g$ for $\widehat{\Omega}_g$:

$$\widehat{\lambda}_D^{\text{unit}} = 1 - q \frac{\left(G^{-1} \sum_{g=1}^G \widehat{S}_g \right)^2}{G^{-1} \sum_{g=1}^G \widehat{S}_g^2}. \quad (13)$$

4 Proof of Reduction to the AAIW Binary Case

We now prove that the continuous weight strictly maps to the AAIW formula in the boolean domain.

Proposition 1. *If $D_{gi} = W_{gi} \in \{0, 1\}$ and covariates Z_{gi} consist strictly of cluster fixed effects, the continuous variation term $\widehat{\Omega}_g$ reduces exactly to the AAIW fixed-effect parameter $\bar{W}_g(1 - \bar{W}_g)$.*

Proof. Because Z_{gi} contains only cluster fixed effects, the FWL residualized treatment simplifies to the within-cluster demeaned treatment:

$$\widetilde{D}_{gi} = W_{gi} - \bar{W}_g.$$

Therefore, our continuous parameter $\widehat{\Omega}_g$ is:

$$\widehat{\Omega}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} (W_{gi} - \bar{W}_g)^2.$$

Expanding the squared binomial yields:

$$\widehat{\Omega}_g = \frac{1}{n_g} \sum_i \left(W_{gi}^2 - 2W_{gi}\bar{W}_g + \bar{W}_g^2 \right).$$

Because W_{gi} is strictly binary, we rely on the fundamental identity $W_{gi}^2 = W_{gi}$. Substituting this identity gives:

$$\begin{aligned} \widehat{\Omega}_g &= \frac{1}{n_g} \sum_i W_{gi} - 2\bar{W}_g \left(\frac{1}{n_g} \sum_i W_{gi} \right) + \bar{W}_g^2 \\ &= \bar{W}_g - 2\bar{W}_g^2 + \bar{W}_g^2 \\ &= \bar{W}_g(1 - \bar{W}_g). \end{aligned}$$

Plugging this directly into the continuous formula establishes the exact equivalence:

$$\hat{\lambda}_D = 1 - q \frac{\left(G^{-1} \sum_g \bar{W}_g (1 - \bar{W}_g) \right)^2}{G^{-1} \sum_g \bar{W}_g^2 (1 - \bar{W}_g)^2},$$

which identically matches the AAIW (2023) fixed-effect CCV weight. Thus, the continuous formula operates as a strict generalization of the original framework. $\square$

5 Proof of Valid Convexity

To ensure $\hat{V}_{CCV,D}$ behaves dynamically without breaking finite sample bounds, we prove $\hat{\lambda}_D$ represents a valid probability measure for interpolation.

Theorem 2. *For any finite continuous treatment distribution where $q \in [0, 1]$ and not all $\hat{\Omega}_g = 0$, $0 \leq \hat{\lambda}_D \leq 1$.*

Proof. Let $\hat{\Omega} \in \mathbb{R}^G$ be a vector of strictly non-negative real numbers representing the cluster variation parameters. By the Cauchy-Schwarz inequality, the inner product of the vector $\hat{\Omega}$ with a vector of ones $\mathbf{1}$ satisfies:

$$\langle \hat{\Omega}, \mathbf{1} \rangle^2 \leq \langle \hat{\Omega}, \hat{\Omega} \rangle \langle \mathbf{1}, \mathbf{1} \rangle.$$

Expanding the sums, this requires:

$$\left(\sum_{g=1}^G \hat{\Omega}_g \right)^2 \leq G \sum_{g=1}^G \hat{\Omega}_g^2.$$

Dividing both sides by G^2 yields the expectation limits:

$$\left(G^{-1} \sum_{g=1}^G \hat{\Omega}_g \right)^2 \leq G^{-1} \sum_{g=1}^G \hat{\Omega}_g^2.$$

Because both the numerator and denominator are strictly non-negative, taking their ratio enforces:

$$0 \leq \frac{\left(G^{-1} \sum_g \hat{\Omega}_g \right)^2}{G^{-1} \sum_g \hat{\Omega}_g^2} \leq 1.$$

Because $q \in [0, 1]$, multiplying the fraction by q preserves the bounds $0 \leq q(\dots) \leq 1$.

Thus, subtracting the term from 1 guarantees:

$$0 \leq 1 - q \frac{\left(G^{-1} \sum_g \hat{\Omega}_g\right)^2}{G^{-1} \sum_g \hat{\Omega}_g^2} \leq 1.$$

Therefore, $\hat{\lambda}_D \in [0, 1]$, confirming $\hat{V}_{CCV,D}$ is a strict convex combination. $\square$

6 Consistency Framework

We formalize the asymptotic limits using standard regularity assumptions under Slutsky's theorem.

Assumption 1. *As the number of clusters grows ($G \rightarrow \infty$), the relative size of any single cluster's contribution remains bounded, such that $\max_g \frac{S_g}{S} \rightarrow 0$. Furthermore, the residualized treatment and errors exhibit bounded fourth moments, and the empirical variance estimators possess non-zero limiting behavior:*

$$\begin{aligned} G^{-1} \sum_g \hat{\Omega}_g &\xrightarrow{p} \Omega_1 > 0, \\ G^{-1} \sum_g \hat{\Omega}_g^2 &\xrightarrow{p} \Omega_2 > 0. \end{aligned}$$

Under Assumption 1, the Continuous Probability Mapping Theorem and Slutsky's Theorem immediately give:

$$\hat{\lambda}_D \xrightarrow{p} \lambda_D = 1 - q \frac{\Omega_1^2}{\Omega_2}.$$

Given that under the standard cluster-level law of large numbers, $\hat{V}_{\text{robust}} \xrightarrow{p} V_{\text{robust}}$ and $\hat{V}_{\text{cluster}} \xrightarrow{p} V_{\text{cluster}}$, we conclude:

$$\hat{V}_{CCV,D} \xrightarrow{p} \lambda_D V_{\text{cluster}} + (1 - \lambda_D) V_{\text{robust}}.$$

7 The Distributional Limit: Higher-Order Moments

The critical caveat in continuous generalization stems from higher-order statistical moments. As established in Proposition 1, a binary treatment enjoys the algebraic property $W^2 = W$, which systematically collapses higher moments. Thus, terms like $E[W^4]$ naturally compress into functions of the mean, allowing the second-moment parameter $\bar{W}_g(1 - \bar{W}_g)$ to fully capture the shape of the design variance.

For an arbitrary continuous treatment D , the optimality of the design variance relies heavily on $E[\tilde{D}^3 | g]$ and $E[\tilde{D}^4 | g]$, which are no longer intrinsically anchored to Ω_g .

Theorem 3 (Location-Scale Sufficiency). *The parameter Ω_g serves as an asymptotically sufficient statistic for the optimal continuous design weight if the treatment assignment process satisfies a common-shape location-scale restriction.*

Proof. Assume the residualized continuous treatment is generated independently given g by:

$$\tilde{D}_{gi} = \sigma_g \epsilon_{gi},$$

where ϵ_{gi} is an identically distributed random error with mean 0, variance 1, and global kurtosis $\kappa < \infty$. Consequently:

$$\hat{\Omega}_g = \frac{1}{n_g} \sum_i \tilde{D}_{gi}^2 \xrightarrow{p} \sigma_g^2.$$

The fourth moment evaluates as:

$$E[\tilde{D}_{gi}^4 | g] = \sigma_g^4 E[\epsilon_{gi}^4] = (\sigma_g^2)^2 \kappa \equiv \Omega_g^2 \kappa.$$

Because the higher moments scale proportionally with Ω_g^2 , Ω_g sufficiently summarizes the cluster-level identifying variation in the assignment process. If kurtosis is heavily heterogeneous across clusters, the CCV estimator remains a computationally valid convex combination but may deviate from strict asymptotic efficiency. $\square$

8 Application: The AEWZ Zero-Bite Boundary

Applying this estimator to the Adverse Effect Wage Rate (AEWR) highlights a desirable empirical property. The treatment variable is left-censored at zero:

$$D_{ct} = \max\{0, \log(\text{AEWR}_{rt}) - \log(w_{ct}^0)\}.$$

In high-wage agricultural counties, w_{ct}^0 chronically exceeds the statutory AEWR_{rt} , resulting in $D_{ct} = 0$ across all time periods t . Consequently, after residualization, $\tilde{D}_{ct} = 0$, yielding $\hat{\Omega}_c = 0$.

If treatment is clustered strictly in a handful of low-wage counties, the proportion of zeros dominates the mean absolute variance. As the sum of squares $G^{-1} \sum \hat{\Omega}_c^2$ remains anchored by the few highly treated clusters, the ratio strictly vanishes:

$$\frac{\left(G^{-1} \sum \hat{\Omega}_c\right)^2}{G^{-1} \sum \hat{\Omega}_c^2} \rightarrow 0 \implies \hat{\lambda}_D \rightarrow 1.$$

Therefore, $\hat{V}_{\text{CCV},D} \rightarrow \hat{V}_{\text{cluster}}$. This boundary limit is highly attractive: if a policy variation is heavily concentrated in a few discrete locations, design-based inference insists that treat-

ment assignment behaves essentially at the macro-level, requiring fully clustered standard errors. The $\hat{\Omega}_g$ extension identifies this boundary and collapses optimally without manual intervention.

References

- [1] Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2023). When Should You Adjust Standard Errors for Clustering? *The Quarterly Journal of Economics*, 138(1), 1–35.